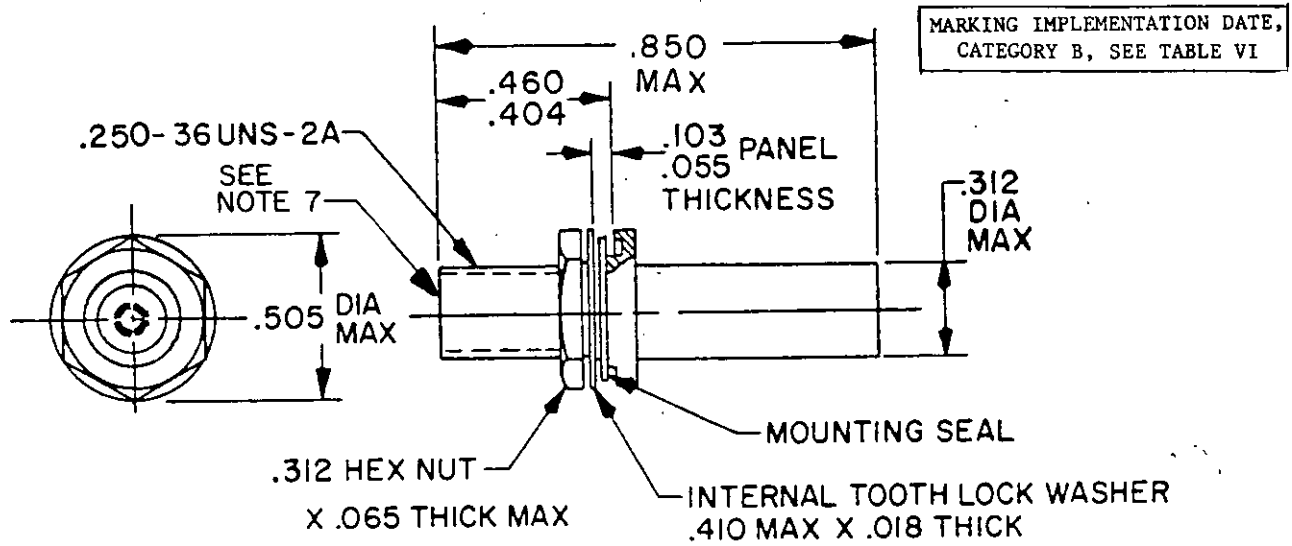
MILITARY SPECIFICATION SHEET

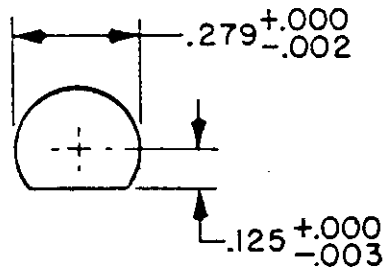
# CONNECTORS, RECEPTACLE, ELECTRICAL, COAXIAL, RADIO FREQUENCY, SERIES SMA, (CABLED, SOCKET, JAM NUT MOUNTED, CLASS 2, SEMIRIGID CABLE)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

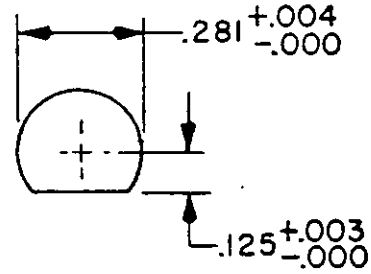
The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-39012.

MARKING IMPLEMENTATION DATE, CATEGORY B, SEE TABLE VI




BODY "D" MOUNTING SURFACE




RECOMMENDED MOUNTING HOLE

| Inches | mm | Inches | mm |
|---|---|---|---|
| .000 | 0.00 | .250 | 6.35 |
| .002 | 0.05 | .279 | 7.09 |
| .003 | 0.08 | .281 | 7.14 |
| .004 | 0.10 | .312 | 7.92 |
| .018 | 0.46 | .404 | 10.26 |
| .055 | 1.40 | .410 | 10.41 |
| .065 | 1.65 | .460 | 11.68 |
| .103 | 2.62 | .505 | 12.83 |
| .125 | 3.18 | .850 | 21.59 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension .505 (12.83 mm) is the largest overall diameter of the connector.
4. Wrench flats to accommodate standard wrench opening per FED-STD-H28, appendix 10.
5. All undimensioned pictorial configurations are for reference purposes only.
6. Dimension .850 (21.59 mm) maximum defines the overall length of connector when assembled to the cable.
7. Series SMA, socket contact interface in accordance with MIL-STD-348.

FIGURE 1. General configuration.


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DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

TABLE I. Dash number and applicable cable.

| Dash no. | Applicable cable 4/ |
|---|---|
| CATEGORY A - FIELD SERVICEABLE (NO SPECIAL TOOLS REQUIRED) 1/ | |
| 3009<br>4009 | M17/133-RG405*<br>M17/133-00001<br>M17/133-00002<br>M17/133-00003<br>M17/133-00004<br>M17/133-00005 |
| 3010<br>4010 | M17/130-RG402*<br>M17/130-00001<br>M17/130-00002<br>M17/130-00003 |
| CATEGORY E - FIELD REPLACEABLE (STANDARD ASSEMBLY TOOL KIT) 2/ 6/ | |
| 3005 5/ | M17/133-RG405*<br>M17/133-00001<br>M17/133-00002<br>M17/133-00003<br>M17/133-00004<br>M17/133-00005 |
| 3006 5/ | M17/130-RG402*<br>M17/130-00001<br>M17/130-00002<br>M17/130-00003 |
| 3007 1/<br>4007 1/ | M17/133-RG405*<br>M17/133-00001<br>M17/133-00002<br>M17/133-00003<br>M17/133-00004<br>M17/133-00005 |
| 3008 1/<br>4008 1/ | M17/130-RG402*<br>M17/130-00001<br>M17/130-00002<br>M17/130-00003 |

TABLE I. Dash number and applicable cable - Continued.

| Dash no. | Applicable cable 4/ | Tool | Positioning dies | Locator pins 3/ | Crimp dies M22520/5 |
|---|---|---|---|---|---|
| CATEGORY F - FIELD REPLACEABLE (MIL-C-22520 CRIMP TOOL) 1/ | | | | | |
| 3207 7/<br>4207 7/ | M17/133-RG405* | M22520/36-01 | M22520/36-02 | M22520/36-05 or M22520/36-17 | --- |
| 3208 7/<br>4208 7/ | M17/130-RG402* | | M22520/36-03 | | |
| 3210<br>4310 | M17/130-RG402* | M22520/5-01 | --- | --- | -05 or -41 cavity B |

See footnotes at end of table.



Source: https://assist.dla.mil -- Downloaded: 202610-21T17:10Z
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* Cable to be used when performing tests requiring cable.

1/ These connectors have captivated center contacts.

2/ Kit number - Ommi Spectra T-200, Amphenol 901-2500, or equivalent.

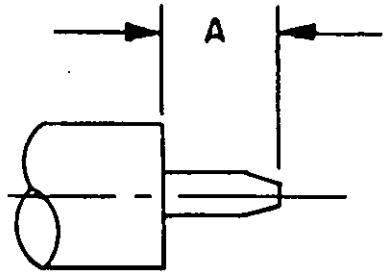
3/ The locators required shall be indicated in the assembly instructions.

4/ MIL-C-17 cables are specified by the basic number. The latest version of each cable shall be applicable.

5/ Inactive for new design. Not for Air Force or Navy use.

6/ All corrosion-resistant steel-bodied connectors shall be gold-plated per MIL-G-45204, type II, class 1.

7/ Not for use in Army equipment.




| CABLE | DIMENSION "A" | |
|---|---|---|
| | MIN | MAX |
| M17/130-RG402<br>M17/130-00001<br>M17/130-00002<br>M17/130-00003 | .080<br>(2.03 mm) | .090<br>(2.29 mm) |
| M17/133-RG405<br>M17/133-00001<br>M17/133-00002<br>M17/133-00003<br>M17/133-00004<br>M17/133-00005 | .065<br>(1.65 mm) | .075<br>(1.91 mm) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tapering of center conductor is optional.

FIGURE 2. Cable stripping dimensions for category E and category F connectors.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:10Z
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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 18 GHz.

Voltage rating: The voltage rating shall be in accordance with table II.

TABLE II. Voltage rating.

| Cable | Voltage max (at sea level) | Voltage max (70,000 ft) |
|---|---|---|
| | V rms | V rms |
| M17/133-RG405<br>M17/133-00001<br>M17/133-00002<br>M17/133-00003<br>M17/133-00004<br>M17/133-00005 | 335 | 85 |
| M17/130-RG402<br>M17/130-00001<br>M17/130-00002<br>M17/130-00003 | 500 | 125 |

Temperature rating: -65°C to +105°C.

Category F: Connectors using semirigid cables with standard stripping dimensions and using standard military assembly tools. The method of assembly of the connector to the cable shall be solderless.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:

Longitudinal force: Not applicable.

Torque: 2 inch-pounds maximum.

Coupling proof torque: Not applicable.

Inspection conditions: For each test of threaded coupling connector where the test is performed on mated pairs, the pairs shall be torqued to 7 to 10 inch-pounds.

Mating characteristics: See MIL-STD-348 and figure 3.

Center contact (socket):

Oversize test pin: .0375 +.0001.

Test pin finish: 16 microinches.

Insertion depth: .030/.045.

Number of insertions: 3.

Insertion force test: Steel test pin diameter .0370 +.0001.

Insertion depth: .050/.075.



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:10Z
Check the source to verify that this is the current version before use.

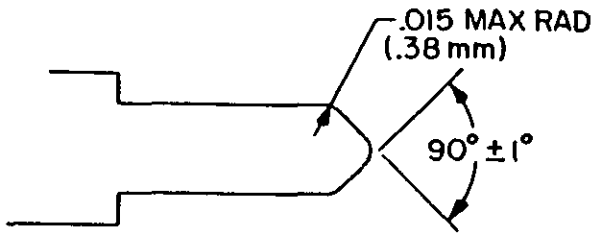



FIGURE 3. Test pin data.

Test pin finish: 16 microinches.

Insertion force: 3 pounds maximum.

Withdrawal force test: Steel test pin diameter .0355 -.0001.

Insertion depth: .050/.075.

Withdrawal force: 1 ounce minimum.

Test pin finish: 16 microinches.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Connectors shall be mounted in mounting hole shown on figure 1 with mating end capped. Test applicable to mounting seal only.

Air pressure: 30 psi.

Duration: 30 seconds minimum, 2 minutes maximum.

Insulation resistance: Method 302, test condition B, MIL-STD-202; 5,000 megohms minimum.

Center contact retention: 6 pounds minimum axial force.

Radial torque: Not applicable.

Corrosion (salt spray): Method 101, test condition B, MIL-STD-202.

Voltage standing wave ratio (VSWR): From 0.5 to 18 GHz, or approximately 80 percent of upper cutoff frequency of the cable, whichever is lower.

| Cables | VSWR | |
|---|---|---|
| | Captive | Non-captive |
| M17/133-RG405<br>M17/133-00001<br>M17/133-00002<br>M17/133-00003<br>M17/133-00004<br>M17/133-00005 | 1.07 +0.01 F (GHz) | 1.07 +0.008 F (GHz) |
| M17/130-RG402<br>M17/130-00001<br>M17/130-00002<br>M17/130-00003 | 1.05 +0.01 F (GHz) | 1.05 +0.008 F (GHz) |



Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:10Z
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Swept frequency VSWR test setup:

Item 6: VSWR shall be less than 1.025 +.002 F (F in GHz).

Item 16: VSWR shall be less than 1.025 +.002 F (F in GHz).

Second step of VSWR checkout procedure: VSWR shall be less than 1.080 +.005 F (F in GHz).

Group B inspection: Use step 5, long cable method.

Qualification and group C inspection: Use step 5, long cable method.

Connector durability:

Insertion and withdrawal force:

500 cycles minimum at 12 cycles per minute maximum.

The connector shall meet mating characteristics and force to engage and disengage requirements.

Contact resistance (in milliohms maximum):

| | Initial | After environment |
|---|---|---|
| Center contact: | 3.0 | 4.0 |
| Outer contact: | 2.0 | Not applicable |
| Outer cable conductor to body: | 0.5 1/ | Not applicable |

Dielectric withstanding voltage at sea level: Method 301 of MIL-STD-202.

| Cables | V rms |
|---|---|
| M17/133-RG405<br>M17/133-00001<br>M17/133-00002<br>M17/133-00003<br>M17/133-00004<br>M17/133-00005 | 1,000 |
| M17/130-RG402<br>M17/130-00001<br>M17/130-00002<br>M17/130-00003 | 1,500 |

Vibration, high frequency: Method 204, test condition D, MIL-STD-202, except the method of mounting shall be approved by the qualifying activity. No discontinuity permitted.

Shock: Method 213, test condition I, MIL-STD-202.

Thermal shock: Method 107, test condition B, MIL-STD-202, except test high temperature shall be +115°C.

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Altitude: 70,000 feet.

1/ Five milliohms is permissible for passivated steel bodied connectors only.



Source: https://assist.dla.mil -- Downloaded: 202640-21T17:10Z
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| Cables | Volts (min) |
|---|---|
| M17/133-RG405<br>M17/133-00001<br>M17/133-00002<br>M17/133-00003<br>M17/133-00004<br>M17/133-00005 | 250 |
| M17/130-RG402<br>M17/130-00001<br>M17/130-00002<br>M17/130-00003 | 375 |

RF high potential withstanding voltage:

Frequency: 5 MHz to 7.5 MHz.

Leakage current: Not applicable.

| Cables | V rms |
|---|---|
| M17/133-RG405<br>M17/133-00001<br>M17/133-00002<br>M17/133-00003<br>M17/133-00004<br>M17/133-00005 | 670 |
| M17/130-RG402<br>M17/130-00001<br>M17/130-00002<br>M17/130-00003 | 1,000 |

Cable retention force: The cable retention force shall be in accordance with table III.

TABLE III. Cable retention force.

| Cable | Pounds (min) | Torque 1/ inch-ounces |
|---|---|---|
| M17/133-RG405<br>M17/133-00001<br>M17/133-00002<br>M17/133-00003<br>M17/133-00004<br>M17/133-00005 | 30 | 16 |
| M17/130-RG402<br>M17/130-00001<br>M17/130-00002<br>M17/130-00003 | 60 | 55 |

1/ Torque to be applied 4 inches maximum from the end of the connector. Reverse bend not applicable.

Coupling mechanism retention force: Not applicable.

RF leakage: -90 dB minimum tested at a frequency between 2 and 3 GHz.

RF insertion loss: dB max = .03 x $\sqrt{\text{freq GHz}}$. Test frequency between 9 and 12.4 GHz.

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Check the source to verify that this is the current version before use.

Supplemental test requirements for qualification and group C inspection for category F connectors.

Six additional connectors shall be selected from the production lot and terminated to produce three cable leads, twelve inches minimum, sixteen inches maximum. The three assemblies shall be subjected to the following tests in sequence: VSWR, thermal shock, VSWR, and cable retention.

The following exceptions to the requirements apply to this supplemental test only:

Voltage standing wave ratio (VSWR) of the cable assembly: From 0.5 to 18 GHz.

| Test cable assemblies | VSWR (initial) | VSWR (after thermal stock) |
|---|---|---|
| M17/133-RG405<br>M17/133-00001<br>M17/133-00002<br>M17/133-00003<br>M17/133-00004<br>M17/133-00005 | 1.15 +.023 F (GHz) | 1.15 +.028 F (GHz) |
| M17/130-RG402<br>M17/130-00001<br>M17/130-00002<br>M17/130-00003 | 1.10 +.023 F (GHz) | 1.10 +.025 F (GHz) |

Thermal shock: 10 cycles.

Part number: M39012/83 (dash number from table I or "B" number from table V).

TABLE IV. Group qualification.

| Group | Submission and qualification of any of the following dash numbers 1/ 2/ | Qualifies the following dash numbers |
|---|---|---|
| I | *02 | *01<br>*02 |
| II | *04 | *03<br>*04 |
| III | *06 | *05<br>*06 |
| IV | *08 | *07<br>*08 |
| V | *10 | *09<br>*10 |
| VI | *208 | *207<br>*208 |
| VII | *210 | *210 |

1/ Individual connectors other than listed are self qualifying only.

2/ Qualification of connectors qualifies connectors of the same material only.

* Body material and connector requirement.



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NOTE: If a connector manufacturer produces a connector that meets all the requirements for two or more connector part numbers (within the same series), the manufacturer may receive qualification approval for two or more connector part numbers qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate part number. For group qualification, the connectors must be of similar design.

TABLE V. Category B-nonfield replaceable (special tools may be required).

NOT FOR AIR FORCE OR NAVY USE. FOR OEM USE ONLY

| M39012/83B 3/ | Applicable cable 2/ |
|---|---|
| 3001 4/ | M17/133-RG405*<br>M17/133-00001<br>M17/133-00002<br>M17/133-00003<br>M17/133-00004<br>M17/133-00005 |
| 3002 4/ | M17/130-RG402*<br>M17/130-00001<br>M17/130-00002<br>M17/130-00003 |
| 3003 1/<br>4003 1/ | M17/133-RG405*<br>M17/133-00001<br>M17/133-00002<br>M17/133-00003<br>M17/133-00004<br>M17/133-00005 |
| 3004 1/<br>4004 1/ | M17/130-RG402*<br>M17/130-00001<br>M17/130-00002<br>M17/130-00003 |

* Cable to be used when performing tests requiring cable.

1/ These connectors have captivated center contacts.

2/ MIL-C-17 cables are specified by the basic number. The latest version of each cable shall be applicable.

3/ All corrosion-resistant steel-bodied connectors whoch are designed to be assembled to the outer conductor using solder shall be gold-plated per MIL-G-45204, type II, class 1.

4/ Inactive for new design.



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TABLE VI. Cross-reference of part numbers.

| Replacement part number M39012/83 1/ | Substitute for part number or type designation |
|---|---|
| B3001 | M39012/83-3001 |
| B3002 | M39012/83-3002 |
| B3003 | M39012/83-3801, -3803, -3003 |
| B4003 | M39012/83-4801, -4803, -4003, -4001 |
| B3004 | M39012/83-3802, -3804, -3004 |
| B4004 | M39012/83-4802, -4804, -4004, -4002 |
| 3007 | M39012/83-3805, -3807 |
| 4007 | M39012/83-4805, -4807, -4005 |
| 3008 | M39012/83-3806, -3808 |
| 4008 | M39012/83-4806, -4808, -4006 |

1/ The new "B" part numbers will be required marking 6 months after the date of this specification. The previous part number may be used in the interim.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AM
Navy - AS, OS, MC
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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